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Dominican Republic Greenhouse Automation



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Motivation

To allow small farms with little resources to gather more information on their crops to produce greater yields.

The solution should be a low-power wireless communication network that sends sensor data from the field to a central, accessible location for a low cost.



Background

For the past 40 years, traditional agriculture has been declining^[1]

- Decline is due to dependencies on supplemental non-farm income for survival^[2]
- Poor Farmers have little land and low production^[3]
- Natural Disasters have greatly harmed the environment^[4]

Farmers in the Dominican Republic rely on their own yields to feed their families and make income



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Frugal Innovation Hub



The Frugal Innovation Hub (FIH) is a SCU program that connects clients with real world problems to students and faculty of the School of Engineering.

Allan Baez Morales, Director of Programs and Partnerships at the FIH, was instrumental in connecting us with IPL and Claudia Diaz. Sean Choi was our advisor for this project. General Engineering majors Henry Kayser and Pedro Alvarenga, along with their advisor Dr. JAK, collaborated with us on our combined final deliverable to IPL.



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Instituto Politécnico Loyola

El Instituto Politécnico Loyola (IPL) is an educational institution which is based in the province of San Cristóbal, Dominican Republic. They have multiple schools for different levels of education, including a technical school, El Instituto Especializado de Estudio Superiores Loyola (IEESL).

Dr. Claudia Diaz, a professor of Telecommunications at IEESL, reached out to the Frugal Innovation Hub to improve upon the communication between the school's greenhouses.



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Testimony from Loyola

“Working with SCU students has been a pleasure, they paid attention to understand the details of the problems we face in the greenhouse and came up with some viable ways to solve them.

Although we were working remotely, they demonstrated good communication skills which made staying in touch a breeze.

I hope that we can have more collaborations between the two institutions in the future.” -Dr. Claudia Diaz



Proposal

Problem Statement

Our objective is to create an automated greenhouse sensor system for Loyola University, with multiple sensors located throughout the greenhouse controlled by a central unit with an interactable application. Due to the cost of implementing and maintaining Wi-Fi and lithium ion batteries, more cost-efficient and eco-friendly solutions are crucial components of this project.

Solution

Use LoRa (Long-Range) communication to receive data on soil temperature and humidity levels, then organizes and displays this information on an interactable application.



Requirements

Functional		Specifications
Safe & reliable data transport		All information transported consistently with no packet loss
Data accessible through interactive application		User given access to information through separate device & UI
Non-functional		
Long-Range	Low Cost	Data Analytics
Energy-efficient	Data Storage/Retrieval	Compact



Use Cases

Loyola University Students/Faculty

- Students and faculty at IPL can monitor their crops' soil and growing conditions from either within or outside the university greenhouses

Dominican Republic Farmers

- Farmers can remotely observe the conditions of their crops, using less labor and growing larger yields

Agricultural Ecologists

- Ecologists can more readily access data on crops in different conditions and locations



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Comparison	LoRa	WiFi	Specifications
Long-range distance communication?	✓	✗	• LoRa: 15-20 km distance range • WiFi: 139 m² area
Reliable? (minimal packet loss)	✓	✓	• LoRa: Some packet loss • WiFi: Minimal packet loss
Energy efficient?	✓	✗	• LoRa: Battery-powered • WiFi: 100-240V (per router)
Cost-effective?	✓	✗	• LoRa: \$10 (per module) • WiFi: \$89+ (per router)



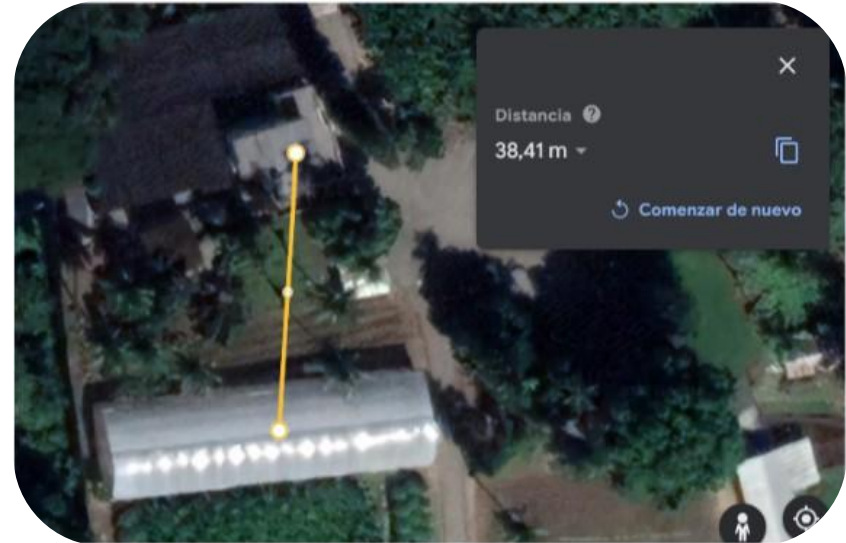
Architecture

Layout

- Node(s) located in greenhouses
 - Reads and sends sensor data
- Gateway listens to node(s)
 - Then transfer data to an interface

Long Range (LoRa)

- Limited bandwidth (Max: 5.5 kbps)
 - Ideal for sensor data
- Radio Module
 - Sending data over radio frequencies





Testbed Design

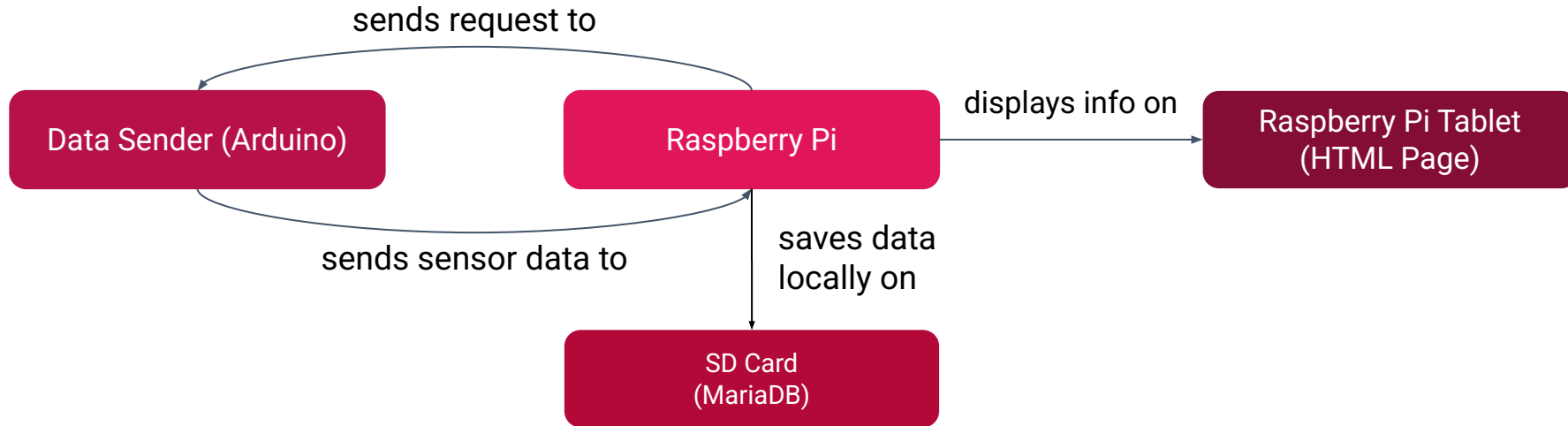


Figure 1. Architecture used for individual testing

Alternate Structures

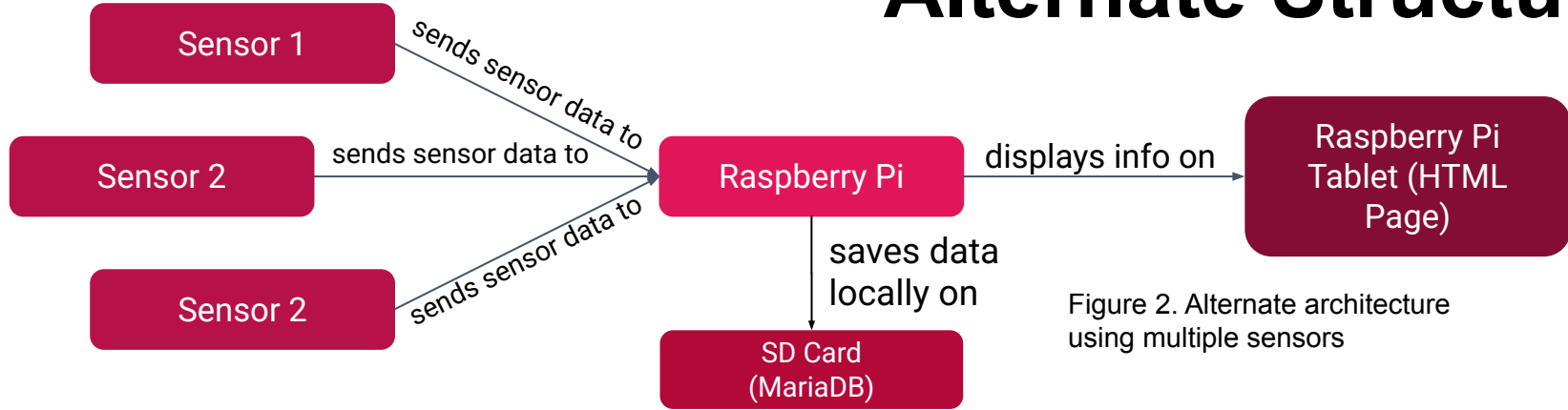


Figure 2. Alternate architecture using multiple sensors

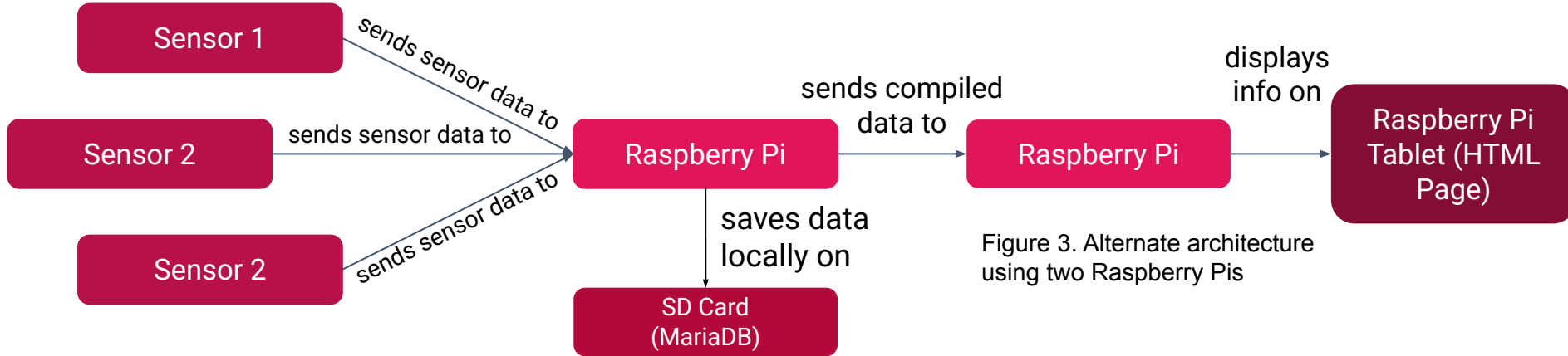


Figure 3. Alternate architecture using two Raspberry Pis



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Tablet Schematic

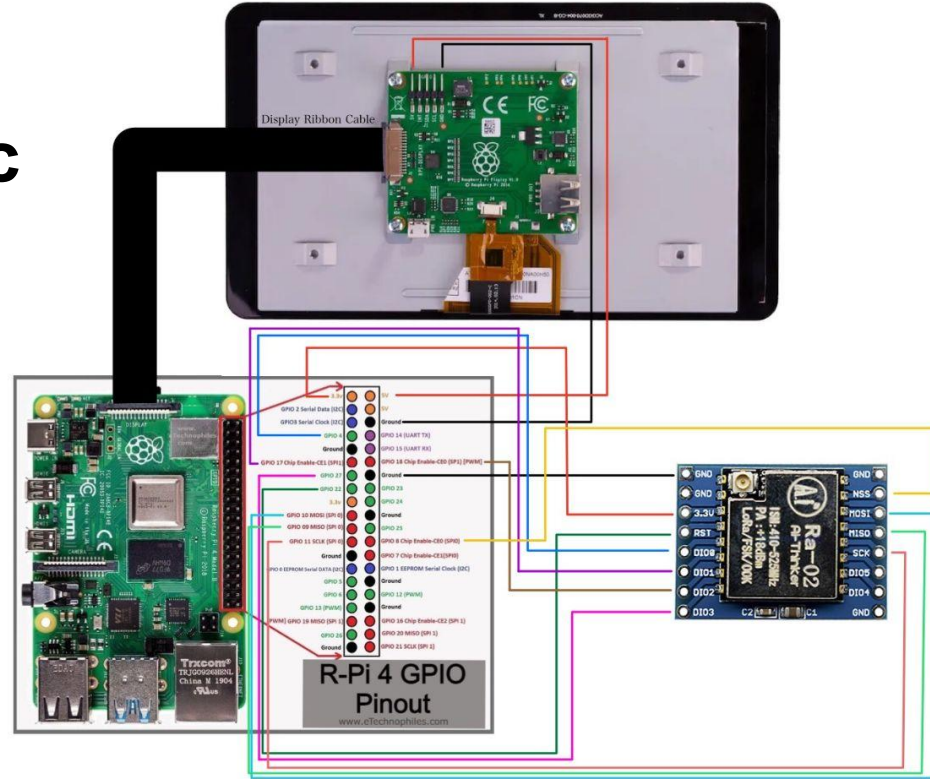


Figure 4. Tablet Schematic

Activity Diagrams

Figure 5. Node Protocol Diagram

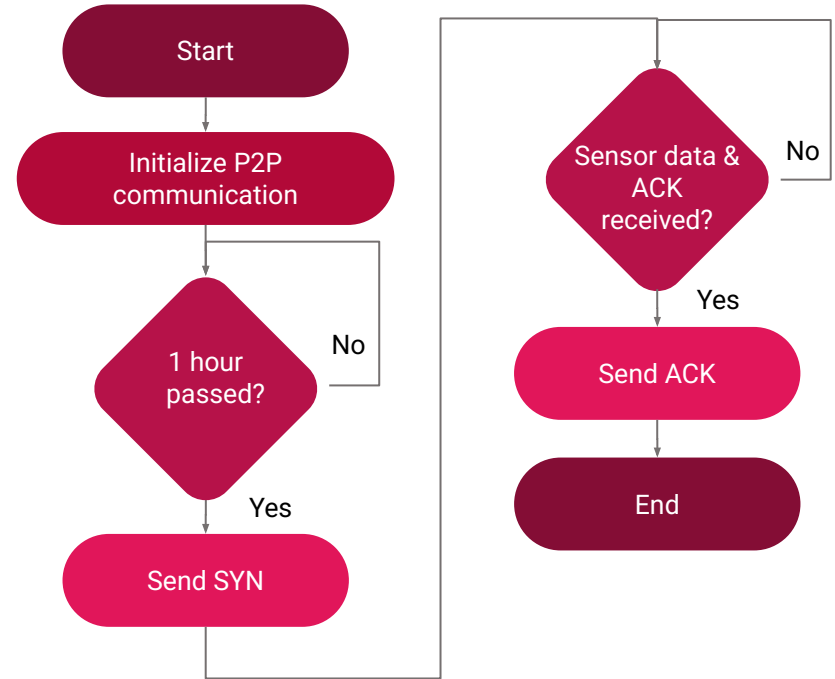
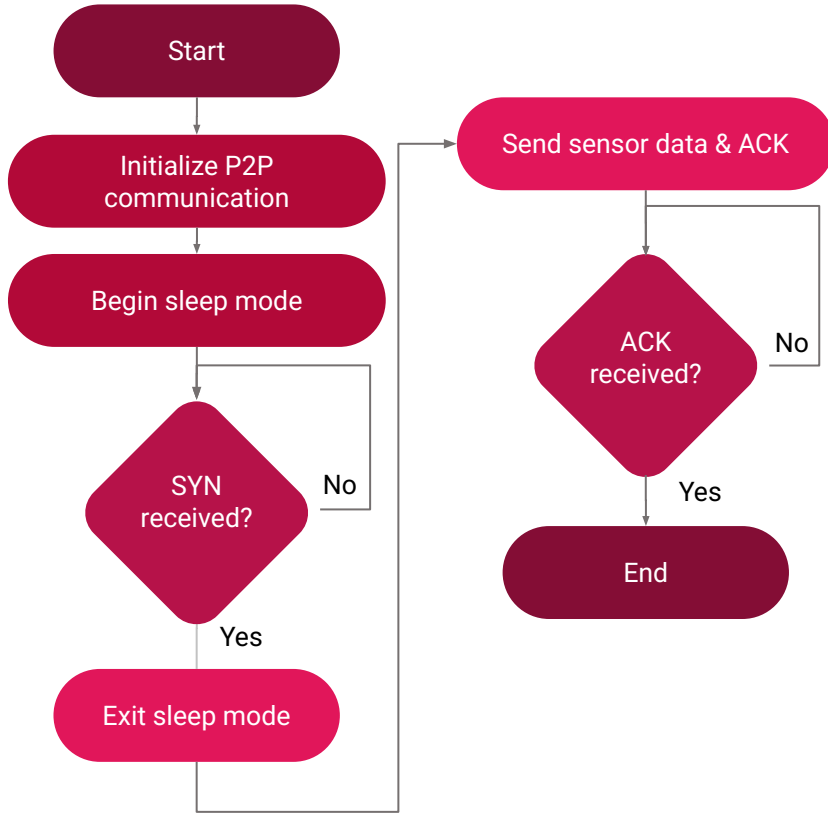


Figure 6. Gateway Protocol Diagram



Application Diagram

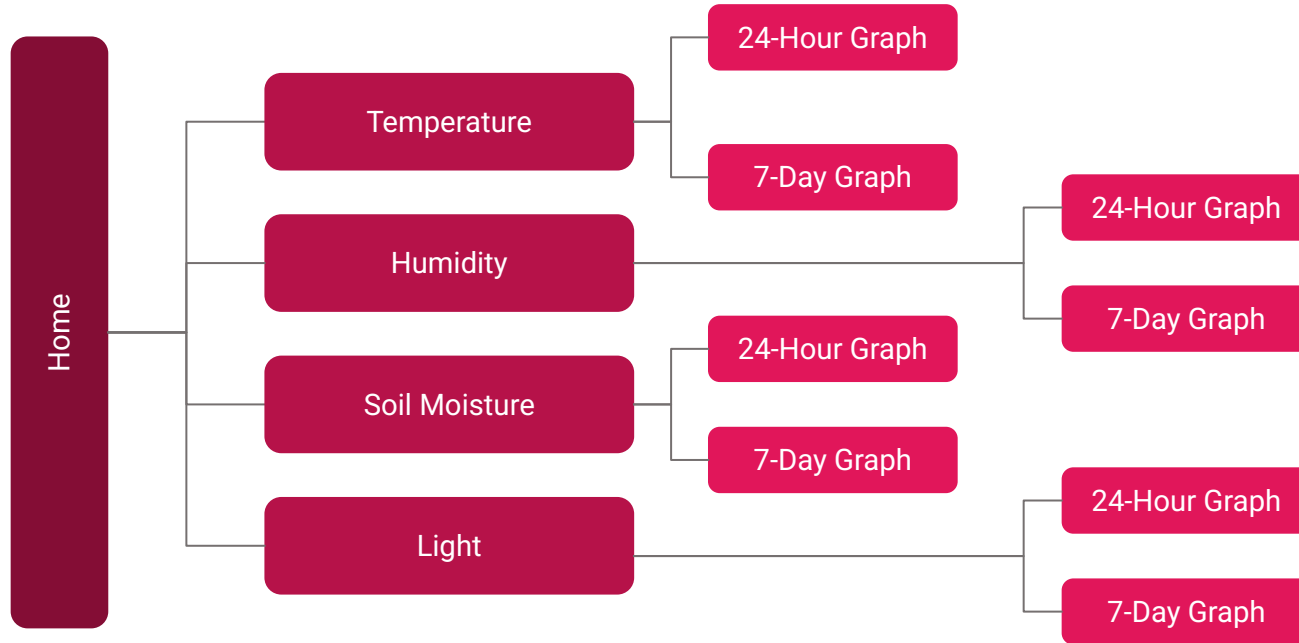


Figure 7. Application Diagram



Technologies Used

Hardware

- Raspberry Pi 4
- Raspberry Pi 7" Touch Screen
- SX1278 LoRa Module
- Arduino Uno (for testing)

Software

- Python
- HTML/CSS/JavaScript
- MySQL

Schedule

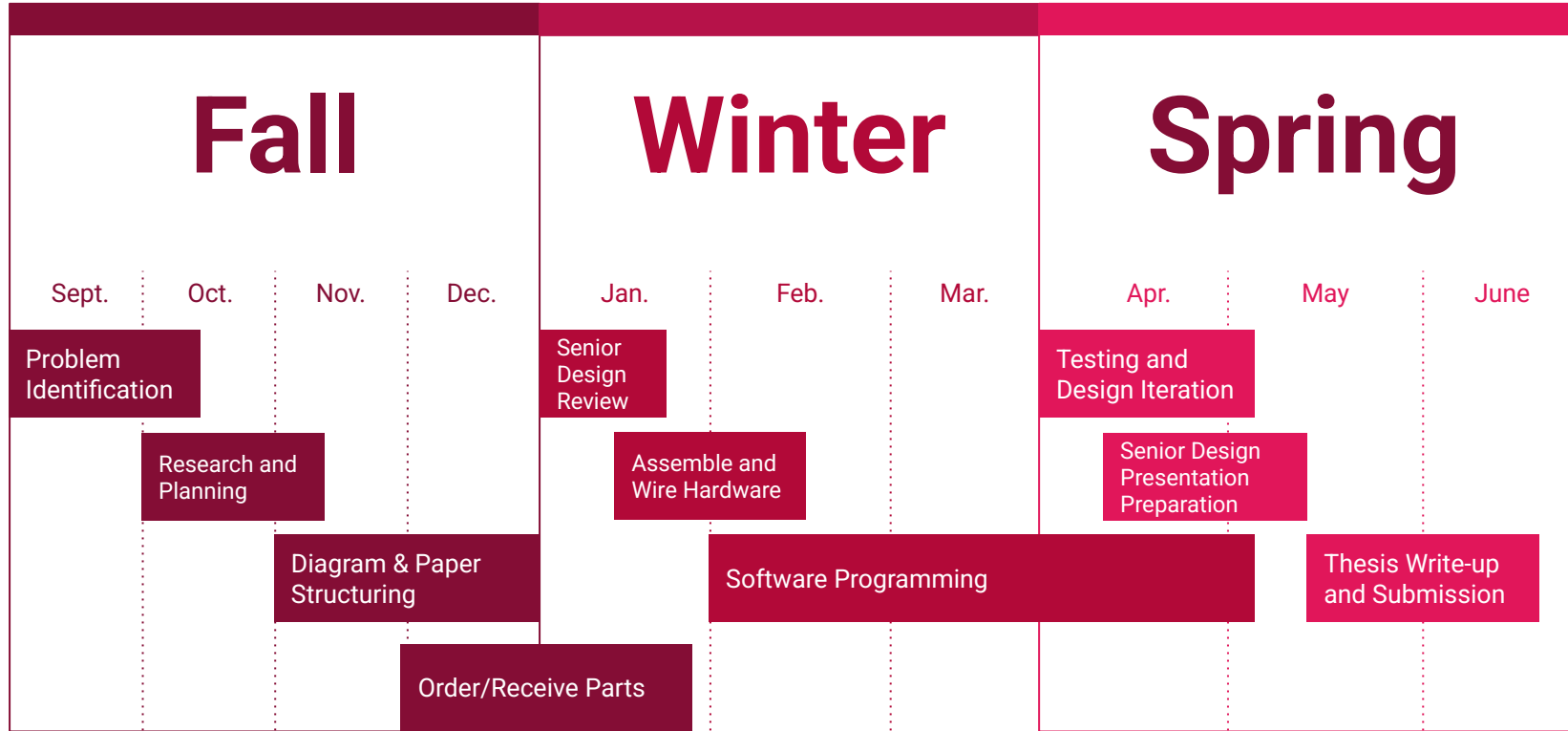


Figure 8. Monthly Timeline



Test Plan

Unit test individual systems

- Data collection
- Data storage
- Application

Test integration between systems

- Data collection and data storage
- Data storage and application

Test system end-to-end

- Real sensor data
- Real-time view of data storage
- Simulated application use



Data Collection

Sending packet: 2

43.00 61.00 6.00 0

Sending packet: 3

48.00 63.00 29.00 4

Sending packet: 4

62.00 14.00 81.00 3

Sending packet: 5

62.00 14.00 81.00 3

Sending packet: 6

74.00 65.00 30.00 0

Received:

43.00 61.00 6.00 0

-92 1 0Waiting for inputs...

Received:

48.00 63.00 29.00 4

-93 1 0Waiting for inputs...

Received:

62.00 14.00 81.00 3

-89 1 0Waiting for inputs...

Received:

62.00 14.00 81.00 3

-91 1 0Waiting for inputs...

Received:

74.00 65.00 30.00 0

-91 1 0Waiting for inputs...█

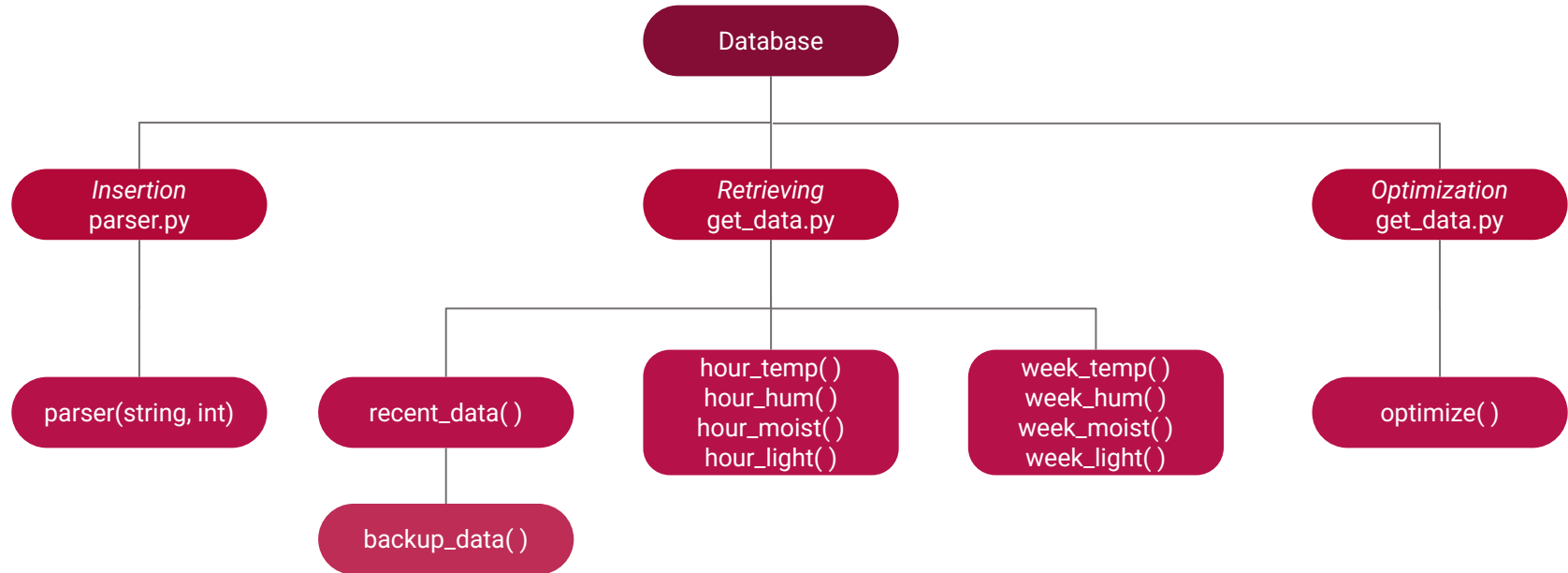


Database Structure

Field	Type	Null	Key	Default	Extra
data_id	int	NO	PRI	NULL	auto_increment
humidity	decimal(3,1)	YES		NULL	
temperature	decimal(3,1)	YES		NULL	
moisture	decimal(3,1)	YES		NULL	
rss_i	int	YES		NULL	
light_intensity	decimal(3,1)	YES		NULL	
time	time	YES		NULL	
date	date	YES		NULL	



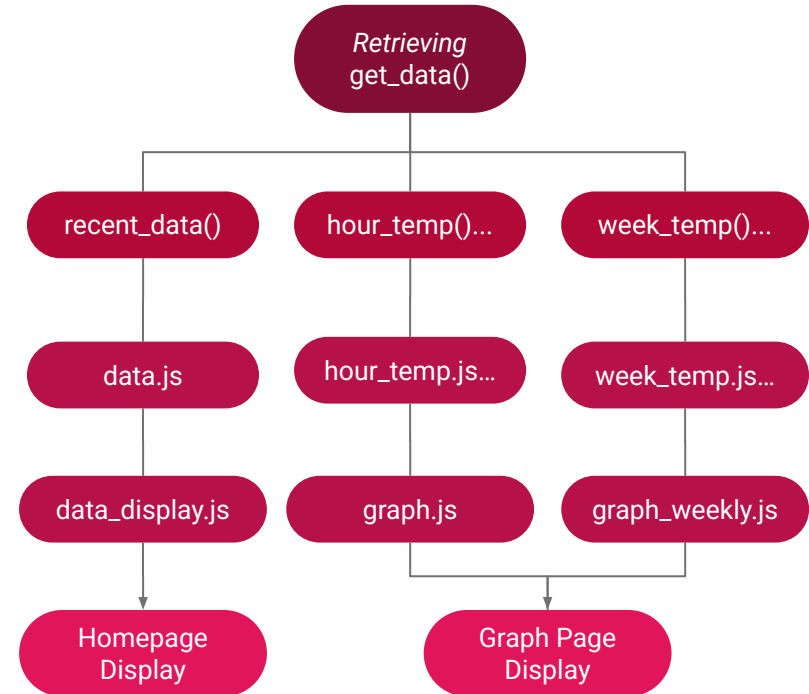
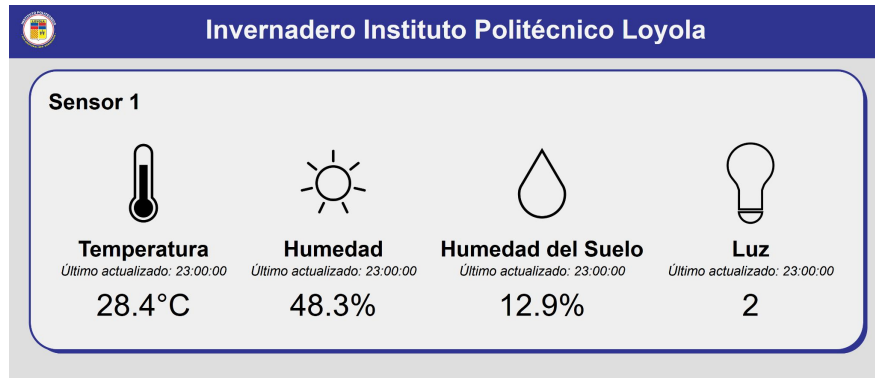
Data Management





Application

- User-friendly interface
- Graphs for trends over 24 hours & 7 days





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Application Demo



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Next Steps for Us

- Finishing the senior design thesis
- Making an instruction manual for the IPL students
- Shipping/delivering solution hardware and software to IPL



Next Steps for the Solution

- Cost analysis of solution for Dominican Republic farmers
 - Current cost: \$380 (compared to \$1000+ for typical solution)
 - Alter hardware to cut costs, keep performance
- All-inclusive solution kits that farmers can understand and use
 - Interface for user-friendly installation
 - Distribution centers for packing and shipping kits



Ethical Considerations

- Benefits to small farms and the economy
 - Agriculture employed 9% of the economically active population in 2019^[5]
 - Farmers have little land and struggle to produce enough food to support their families^[5]
- Low power consumption
 - Implementing LoRa vs Wifi
 - Solar powered batteries



Further References

Loyola University Website

<https://superior.ipl.edu.do>

MariaDB Resources

<https://linuxhint.com/install-setup-mariadb-server-raspberry-pi/>

<https://raspberrypi.com/install-mariadb-raspberry-pi/>

<https://www.digitalocean.com/community/tutorials/how-to-store-and-retrieve-data-in-mariadb-using-python-on-ubuntu-18-04>

Raspberry Pi

<https://howchoo.com/pi/raspberry-pi-touchscreen-setup#:~:text=Connect%20the%20jumper%20cables%20to%20power%20the%20display,-There%20are%20a&text=The%20kit%20comes%20with%20four,pin%20on%20the%20display%20board>

<https://www.raspberrypi.com/software/>

LoRa Communication

<https://github.com/sandeepmistry/arduino-LoRa>

<https://www.metageek.com/training/resources/understanding-rssi/>

<https://iotdesignpro.com/projects/wireless-communication-between-arduino-and-raspberry-pi-using-lora-module-sx1278>

<https://circuitdigest.com/microcontroller-projects/raspberry-pi-with-lora-peer-to-peer-communication-with-arduino>

https://www.researchgate.net/publication/337365464_Development_of_environmental_monitoring_systems_based_on_LoRa_with_cloud_integration_for_rural_area

<https://zerynth.com/blog/lora-nodes-real-time-data-plotting-using-python-for-microcontrollers/>

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<https://www.electronics-lab.com/project/introduction-lora-send-data-two-arduino-using-lora/>





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Thank You

